# Block-Level Urban Growth Prediction from Google Open Buildings Temporal Data

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***Abstract*—We use the Google Open Buildings 2.5D Temporal dataset [1] to predict where new construction will occur in 25 cities across Africa, South Asia, South-East Asia and Latin America. Rather than predict at the noisy pixel level, we aggregate to 17 m × 17 m blocks (4×4 pixels), which removes most isolated-pixel detector noise while keeping building-level spatial resolution. Three neural models are compared against a mean-rate baseline on 500 patches sampled uniformly at random from five fully held-out test cities. A Block U-Net that ingests seven years of building presence history reaches PR-AUC 0.662 and lift 2.39×: its top-ranked 17 m blocks contain new construction roughly 2.4 times more often than a random selection of equal area. Adding road proximity and city-centre distance as extra input channels produces no measurable improvement, suggesting the seven-year temporal stack already contains this information.**

## I. Introduction

Anticipating urban growth at the block level is useful for planning roads, schools, water and waste infrastructure before construction outpaces capacity. In much of the Global South authoritative land-use registries are incomplete or out of date, so building-scale satellite-derived datasets are the most practical source of ground truth. Google's Open Buildings 2.5D Temporal dataset [1] publishes yearly per-pixel building presence rasters at roughly 4 m effective resolution from 2016 onward, covering most of the developing world.

In this work we use seven years of building presence history (2016–2022) to predict where new construction will appear in 2023. Rather than work at the noisy pixel level we aggregate to 17 m × 17 m blocks and rank each block by its predicted growth rate. Top-ranked blocks are the model's growth hotspots, and the practical question is how much more often those top-ranked blocks contain new construction than a random selection of equal area.

We compare three neural architectures (a global Patch CNN, a Block U-Net, and a Block U-Net with road and centre features) against a mean-rate baseline. Evaluation uses 500 patches sampled uniformly at random from five fully held-out test cities, so test pixels are never seen during training. We also test whether road proximity and city-centre distance, both motivated by recent literature [4], [5], improve the model when the seven-year temporal stack is already present.

## II. Data

We use the building presence band of the Google Open Buildings 2.5D Temporal dataset [1], which provides annual rasters from 2016 to 2023 with one channel giving the model's confidence that the pixel is covered by a building in that year. Building presence is derived by a teacher–student pipeline in which a high-resolution detector trained on commercial imagery distils its predictions into a student model that runs on freely available Sentinel-2 imagery [1], allowing the resulting dataset to scale continent-wide.

### A. City selection and splits

Twenty-five cities are sampled across Africa, South Asia, South-East Asia and Latin America to cover a wide range of growth patterns, climates and urban forms. Each city is centred on its geographic centre point and a $0.04^\circ \times 0.04^\circ$ window is read for each year, resampled to a $1024 \times 1024$ raster. Cities are split by city, not by patch, into 15 training, 5 validation and 5 test cities (Table I). This split ensures that no spatial pixel from a test city is ever seen during training, which is a stricter test than randomly splitting patches within the same city.

TABLE I
City splits.

| Split | Cities |
|---|---|
| Train | new_cairo, accra, nairobi, lagos, addis_ababa, dhaka, delhi, karachi, chennai, jakarta, ho_chi_minh, lima, bogota, quito, medellin |
| Val | manila, phnom_penh, kampala, kigali, bangkok |
| Test | mexico_city, sao_paulo, dar_es_salaam, lahore, hanoi |

### B. Training patches

From each city's $1024 \times 1024$ raster we sample $128 \times 128$ pixel patches (roughly $552 \times 552$ m on the ground) using a 70/30 mix of positives (at least 0.2% of currently-empty pixels become built in 2022→2023) and negatives (no significant growth but at least 2% of pixels already built in 2022). The training and validation patches are growth-biased on purpose so the model sees a useful number of positive examples; final evaluation uses an unbiased uniform sample (Section IV). The sampler ran across all 25 cities and produced 2,472 patches: 1,542 training and 480 validation, plus 450 held-back patches we did not use in the reported evaluation.

### C. From pixel labels to 17 m blocks

Working at the raw pixel level is problematic because the source labels are noisy: a manual inspection of all pixels marked as "new building" for the 2022→2023 transition found that the majority are isolated single-pixel detections rather than real building footprints. The underlying Sentinel-2 input is 10 m, and the published Open Buildings rasters are upsampled to roughly 4 m [1], so a real new building should correspond to a small connected cluster rather than a single pixel.

To remove this noise without losing useful resolution, we aggregate each $128 \times 128$ patch into a $32 \times 32$ grid of 17 m blocks (each block is a $4 \times 4$ pixel cell). The block label is the fraction of its 16 pixels that become newly built in 2022→2023, in $[0, 1]$. A typical building footprint spans 1–3 blocks, so block-level supervision is still building-scale but is much less sensitive to per-pixel flicker. The mean training block growth rate at this scale is 3.16%.

### D. Auxiliary channels

Two external sources are added for the final model:

- **Road proximity.** Drivable road network for each city is downloaded via OSMnx [3] and rasterised onto the same grid as the presence maps. For each pixel we compute the Euclidean distance to the nearest road and convert it to a proximity score $1/(1+d/300\text{m})$, near 1 next to a road and decaying smoothly with distance. The choice is motivated by macro-economic evidence that within-city road layout predicts urban growth in Sub-Saharan Africa [4].
- **City-centre distance.** A per-pixel normalised radial distance from the AOI centre. Urban-growth models across African countries include this kind of variable as a coarse signal for "core vs. fringe" position [5].

## III. Methods

### A. Problem formulation

Let $B_t \in [0, 1]^{H \times W}$ be the building presence confidence map of a patch at year $t$. The pixel-level growth indicator is $g_{t \to t+1}(i, j) = \mathbb{1}[B_t(i, j) < 0.5] \cdot \mathbb{1}[B_{t+1}(i, j) \geq 0.5]$, i.e. the pixel was empty at year $t$ and is built at year $t+1$. The block-level target is the mean of $g$ over each $4 \times 4$ block. The model outputs a $32 \times 32$ block-level score map, optimised against the block target with Huber loss [8] (delta = 0.02), which is more stable than $\ell_2$ on near-zero targets and less sensitive than $\ell_1$ to small noise.

### B. Models

Four configurations are compared (Table II).

**Mean baseline.** A single constant equal to the mean training block growth rate, broadcast over every block. This sets the lift = 1.0 reference point.

**Patch CNN.** A small convolutional encoder followed by global average pooling and a two-layer MLP, producing one scalar per patch. The same scalar is broadcast to every block in the patch at evaluation time. This isolates the value of the model's *global* growth-rate estimate from any within-patch spatial detail.

TABLE II
MODELS COMPARED. "PER BLOCK" MEANS THE MODEL PRODUCES A DISTINCT SCORE FOR EVERY 17 M BLOCK IN THE PATCH; "PER PATCH" MEANS A SINGLE SCALAR SHARED BY ALL 1024 BLOCKS.

| Model | Output | Parameters |
|---|---|---|
| Mean baseline | constant per patch | — |
| Patch CNN | one scalar per patch | 166k |
| Block U-Net | per block | 1.57M |
| Block U-Net + roads/centre | per block | 1.57M |

**Block U-Net.** A U-Net [2] encoder–decoder with four pooling levels (encoder widths 32-64-128-128-128). The output is a $32 \times 32$ logit map at block resolution, passed through a sigmoid. Input is a 7-channel stack of building presence from 2016 to 2022.

**Block U-Net + roads/centre.** Same architecture and training recipe as Block U-Net, but with two extra input channels: road proximity and normalised distance from the city centre, both described in Section II-D. This isolates the value of those auxiliary features when seven years of temporal history are already available.

### C. Training

All neural models are trained with Adam [6] at learning rate $10^{-3}$, batch size 16, for up to 50 epochs with early stopping (patience 7) on validation Huber loss. Learning rate is halved on validation plateau. Training is deterministic via a fixed seed and the model checkpoints are cached to disk, so once a model has been trained the experiment can be reloaded instantly.

## IV. Evaluation

### A. Unbiased test set

The patches used for training and validation are growth-biased on purpose. To estimate realistic deployment quality we build a separate unbiased test set: 500 patches sampled uniformly at random from the five held-out test cities, 100 per city, with no growth filter at all. This represents the situation a planner would face: the model is applied to arbitrary parts of the city, most of which have little or no new construction. On this test set 27.7% of blocks see any new construction in 2022→2023 (this is the per-block positive rate), and the per-pixel growth rate is 3.63%.

### B. Metrics

**PR-AUC** (primary). Area under the per-block precision–recall curve. PR-AUC is threshold-free and well-suited to imbalanced binary tasks [7].

**Lift.** PR-AUC divided by the block positive rate (27.7%). Lift = 1.0 corresponds to random ranking; lift = 2.4 means the model's top-ranked blocks contain new construction 2.4 times more often than a random selection of equal area. We use lift as the headline metric because it has a direct planning interpretation.

**$R^2$**. Variance in block growth rates explained by the model. The mean baseline scores 0 by construction.

**MAE.** Mean absolute prediction error in percentage points of growth rate per block.

**F1.** Binary F1 at the 6.25% block threshold (i.e. a block counts as positive if at least one of its 16 pixels grew). All neural models are scored at the same threshold.

## V. Results

### A. Main comparison

Table III reports the main comparison. The mean baseline scores PR-AUC 0.277, exactly the block positive rate, by construction. The Patch CNN improves PR-AUC to 0.408 (lift 1.47×). The Block U-Net is the clear winner at PR-AUC 0.662 (lift 2.39×): its top-ranked 17 m blocks contain actual new construction nearly two and a half times more often than a random selection of equal area. Adding road proximity and city-centre distance as extra input channels does not move the needle (PR-AUC 0.658, lift 2.38×).

### B. Why the Patch CNN underperforms the Block U-Net

The Patch CNN predicts only *how much* a $552 \times 552$ m patch grows in total; it assigns the same score to every block inside that patch. The Block U-Net resolves *where* within the patch growth concentrates. Even when the two models agree on the total growth rate of a neighbourhood, the Block U-Net captures useful within-patch variation that the Patch CNN cannot represent. This is reflected in the regression metrics: $R^2$ improves from 0.032 to 0.247 and MAE drops from 5.10 to 3.28 percentage points, while the ranking-quality gap (PR-AUC) is also large.

### C. Why roads and city-centre distance add nothing here

The +roads/centre model achieves essentially the same PR-AUC and lift as the temporal-only Block U-Net. We think the reason is that seven years of building presence history already encodes the same information these auxiliary features carry. The model can read off, from $B_{2016}$ through $B_{2022}$ alone, where roads sit (they appear as linear gaps between built clusters) and where the city centre is (the dense built-up core). Adding road proximity and centre distance as separate channels gives the model information it has already inferred internally.

This contradicts what we expected from the literature on road-driven urban growth in Sub-Saharan Africa [4] and from the standard practice of including centre-distance variables in continent-scale urban expansion models [5]. Those features matter when the model has no other view of the city. Here the temporal stack provides a richer view, and the extra channels become redundant.

### D. The label-noise ceiling

All neural variants on this test set converge around lift 2.4×. We attribute this ceiling to per-pixel detection noise in the source labels: many "new building" pixels in 2022→2023 are isolated detector flicker rather than true construction. Aggregating to 17 m blocks reduces but does not eliminate this effect, because some blocks contain only one or two noisy pixels and still flip from 0 to a small positive fraction. Pushing performance materially higher would require either a cleaner ground-truth source (multi-year persistence filtering, weak labels at the building level rather than the pixel level, or manually verified construction events) or a multi-year forecasting horizon that allows the model to integrate evidence across more than one transition.

## VI. Conclusion

A Block U-Net trained on seven years of Open Buildings Temporal presence history achieves PR-AUC 0.662 and lift 2.39× on a 500-patch unbiased citywide test set drawn from five held-out cities. The block-level output (17 m grid) is the right operating scale for this dataset: it removes pixel-level detector noise while remaining building-scale. Adding road proximity and city-centre distance as auxiliary channels did not help once the temporal stack was in place, suggesting that what the model needs most is more years of clean history rather than more spatial side-information.

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TABLE III
PERFORMANCE ON THE 500-PATCH UNBIASED TEST SET (2022→2023, 5 HELD-OUT CITIES). BLOCK POSITIVE RATE IS 27.7%, SO LIFT = PR-AUC / 0.277. LIFT = 1.0 IS THE RANDOM-RANKING FLOOR.

| Model | PR-AUC | Lift | $R^2$ | MAE %pt | F1 |
|---|---|---|---|---|---|
| Mean baseline | 0.277 | 1.00 | −0.003 | 5.04 | 0.000 |
| Patch CNN | 0.408 | 1.47 | 0.032 | 5.10 | 0.256 |
| Block U-Net | **0.662** | **2.39** | **0.247** | 3.28 | **0.493** |
| Block U-Net + roads/centre | 0.658 | 2.38 | 0.227 | **3.26** | 0.462 |